FMTH0301/Rev.5.3

Course Plan

Semester: III Year: 2024-25

Course Title: Computer Organization and Architecture	Course Code :21ECSC201
Total Contact hours:68 hours.	Duration of ESA :3 hours
ISA Marks: 50	ESA Marks:100
Lesson Plan Author: Aruna S. Nayak, Namrata D. Hiremath, Umadevi F.M.	Date: 10.09.2024
Checked By: Dr. Vijayalakshmi M.	Date: 11.09.2024

Brief description of the course:

The Computer Organization & Architecture course covers the core principles of computer systems, including memory hierarchy, input/output mechanisms, interrupts, pipelining and the basics of parallel processing. It offers a thorough understanding of how modern computer systems operate and interconnect at both hardware and software levels. By bridging the gap between hardware and software, this course equips future engineers with the skills to design efficient, scalable and robust computing systems. A key component of the course is a hands-on activity where students design a simple processor using a simulation tool, providing them with valuable insights into the instruction lifecycle and the internal workings of processors.

Prerequisites: The concepts on Digital electronics, Boolean Algebra, combinational and sequential circuits from Basic Electronics course.

Course Outcomes (COs):

At the end of the course the student should be able to:

1. Acquire basic knowledge in computer fundamentals, performance analysis and interconnection principles in modern computing.
2. Build memory modules to perform I/O operations.
3. Apply Instruction Set Architecture principles to effectively design and implement datapath in computer systems.
4. Explain the concept of parallelism with reference to computer performance.
5. Integrate and simulate all modules to develop a complete processor, followed by a comprehensive performance analysis.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

Course Title: **Computer Organization and Architecture**	Semester: **III**
Course Code: **21ECSC201**	Year: **2024-25**

Course Outcomes (COs)/ Program Outcomes (POs	**Program Outcome**												**PSO**	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
1. Acquire basic knowledge in computer fundamentals, performance analysis and interconnection principles in modern computing	H													
2. Build memory modules to perform I/O operations.	H				M									
3. Apply Instruction Set Architecture principles to effectively design and implement datapath in computer systems	H	H			M									
4. Explain the concept of parallelism with reference to computer performance.	M	M												
5. Integrate and simulate all modules to develop a complete processor, followed by a comprehensive performance analysis.		M	M		M				M					

Degree of compliance **L**: Low **M**: Medium **H**: High

Competency addressed in the Course and corresponding Performance Indicators

Competency	Performance Indicators
1.4: Demonstrate competence in computer science engineering knowledge.	**1.4.4:**Apply machine dependent/independent features to build system modules.
2.1: Demonstrate an ability to identify and characterize an engineering problem.	**2.1.2:**Identify processes, modules, variables, and parameters of computer based system to solve the problems.
3.3: Demonstrate an ability to select the optimal design scheme for further development.	**3.3.1:**Apply suitable criteria to select optimal design solutions for further development.
5.3: Demonstrate an ability to apply IT tools for the chosen engineering activity	**5.3.1:**Demonstrate proficiency in using IT tools for performing engineering activity
9.3: Demonstrate success in a team-based project	**9.3.1:** Present results as a team, with smooth integration of contributions from all individual efforts

Eg:**1.2.3**: Represents ProgramOutcome-1, Competency-2, andPerformanceIndicators-3.

Course Content

Course Code: **21ECSC201**	Course Title: **Computer Organization and Architecture**	
L-T-P : 3-0-1	Credits: 4	Contact Hours: 5 hrs/week
ISA Marks: 50	ESA Marks: 50	Total Marks: 100
Teaching Hours: 40	Practical:28	Exam Duration: 3 hours

Unit - I	
Chapter 1: Computer Fundamentals **Basic Concepts and Computer Evolution**: Organization and Architecture, Structure and Function, A Brief History of Computers, The Evolution of the Intel x86 Architecture, Embedded Systems **Performance Issues:** Two Laws that Provide Insight: Amdahl's Law and Little's Law, Basic Measures of Computer Performance, Calculating the Mean, Benchmarks and Spec. **A Top-Level View of Computer Function and Interconnection:** Computer Components, Computer Function, Interconnection Structures, Bus Interconnection, Point-to-Point Interconnect	04 Hours
Chapter 2: Computer System **Memory:** Computer Memory System Overview, Cache Memory Principles, Elements of Cache Design, Semiconductor Main Memory, DDR DRAM **Input/Output**: External Devices, I/O Modules, Programmed I/O, Interrupt-Driven I/O, Direct Memory Access	06 Hours
Chapter 3: The Central Processing Unit **Instruction Sets:**Characteristics and Functions:Machine Instruction Characteristics, Types of Operands, Types of Operations **Addressing Modes and Formats:** Addressing Modes, Instruction Formats, Assembly Language	06 Hours
Unit – II	
Chapter 4: The Processor **Processor Structure and Function:** Processor Organization, Register Organization, Instruction Cycle, Instruction Pipelining **Instruction-Level Parallelism and Superscalar Processors:**Overview, Design Issues	08 Hours
Chapter 5: Control Unit Operation & Operating System support	08 Hours

Micro-Operations, Control of the Processor, Operating system overview, scheduling & memory management	
Unit – III	
Chapter 6: Parallel Organization **Parallel Processing:** Multiple Processor Organizations, Symmetric Multiprocessors, Cache Coherence and the MESI Protocol, Multithreading and Chip Multiprocessors **Multicore Computers:** Hardware Performance Issues, Software Performance Issues, Multicore Organization, Heterogeneous Multicore Organization.	04 Hours
Chapter 7: General-Purpose Graphic Processing Units CUDA Basics, GPU versus CPU, GPU Architecture Overview	04 Hours

Text Books (List of books as mentioned in the approved syllabus)

1. William Stallings, Computer Organization and Architecture Designing for Performance, 10th Ed, Pearson Education, 2020.

References

1. Hamacher C., Vranesic Z., and Zaky S., Manjikian N., Computer Organization and Embedded Systems, 6Ed, McGraw Hill, 2020.
2. John L. Hennessy and David A. Patterson, Computer Architecture: A Quantitative Approach 5th Edition, Elsevier publication, 2017.
3. Kai Hwang, Advanced Computer Architecture Parallelism Scalability Programmability, Tata McGraw Hill 2008

Evaluation Scheme

In-Semester Assessment Scheme

Assessment	Conducted for marks	Weightage in Marks
ISA-1 (Theory)	40	37.5
ISA-2 (Theory)	40	
ISA-1 (Lab)	30	12.5
ISA-2 (Lab)	30	
	Total	**50**

End-Semester Assessment Scheme

Assessment	Conducted for marks	Weightage in Marks
Theory	100	37.5
Lab	40	12.5
	Total	**50**

Experiment wise plan

List of Experiments/Jobs planned to meet the requirements of the course

Expt/ Job No.	Experiment/ Job details	No. of Lab sessions/batch
1.	Logisim Tool Demo	01
2.	Combinational Circuits (Half Adder, Full Adder, Decoder, Multiplexer)	01
3.	Building ALU	01
4.	In Semester Assessment #1	01
5.	1-bit RAM Cell and building bigger RAM	02
6.	Design and simulation of main memory organization	02
7.	Design and simulation of register organization	02
8.	Design and simulation of datapath for processor design.	03
9.	In Semester Assessment #2	01

In Semester Assessment#1 (ISA) Rubrics for Practical (P) component

ISA evaluation marks: 30 (5th Week of COE)									
Sl. No.	**Parameters**	**Excellent**	**Very Good**	**Good**	**Average**	**Scope for improvement**	**BL**	**CO**	**PI**
1.	Identifying components with its functionality. **(05 M)**	Accurately identifies all components and demonstrates thorough understanding of all functionalities **(5 M)**	Correctly identifies nearly all components and shows clear understanding with minor gaps. **(4 M)**	Identifies most components correctly and demonstrates a general understanding, but misses some details. **(3 M)**	Identifies some components correctly and shows basic understanding. **(2 M)**	Fails to identify most components with no understanding of functionality. **(1-0 M)**	L3	CO 1	2.1.2
2.	Design of processor components **(15 M)**	The solution demonstrates clarity in component design, with a clear understanding. **(15-13M)**	The solution demonstrates component design understanding of each module. **(12-10 M)**	The solution demonstrates component design with a moderate understanding. **(9-6M)**	The solution presents a component design but lacks a comprehensive understanding. **(5-3M)**	The solution lacks clarity showing a basic approach. **(2-0M)**	L3	CO 3	3.3.1
3.	Simulation of the processor components **(10 M)**	Circuit is simulated successfully, and all outputs match expected results without any errors. **(10-9M)**	Simulation produces mostly correct outputs with minor errors. **(8-7 M)**	Simulation produces correct outputs for most test cases, but noticeable errors remain. **(6-5M)**	Simulation shows errors in key outputs, though some parts function as expected. **(4-3M)**	Simulation fails or produces incorrect results for most test cases. **(2-0 M)**	L3	CO 5	5.3.1

In Semester Assessment# 2 (ISA) Rubrics for Practical (P) component

ISA evaluation marks: 30 (12th week of COE)									
Sl. No.	**Parameters**	**Excellent**	**Very Good**	**Good**	**Average**	**Scope for improvement**	**BL**	**CO**	**PI**
1.	Identifying components with its functionality. **(5M)**	Accurately identifies all components and demonstrates thorough understanding of all functionalities **(5 M)**	Correctly identifies nearly all components and shows clear understanding with minor gaps. **(4 M)**	Identifies most components correctly and demonstrates a general understanding, but misses some details. **(3 M)**	Identifies some components correctly and shows basic understanding. **(2 M)**	Fails to identify most components with no understanding of functionality. **(1-0 M)**	L3	CO 1	2.1.2
2.	Design of datapath **(10M)**	The solution demonstrates exceptional innovation in datapath design, with a clear understanding of each module. **(10-9M)**	The solution demonstrates datapath design, with a clear understanding of each module. **(8-7M)**	The solution demonstrates datapath design, with a moderate understanding of each module. **(6-5M)**	The solution presents a datapath design but lacks a comprehensive understanding of the functionality and interactions of each individual module. **(4-3M)**	The solution lacks creativity, showing a basic approach. **(2-0 M)**	L3	CO 3	3.3.1
3.	Implementation **(5 M)**	The datapath is implemented with all the specifications provided in the problem statement with own innovative thinking. **(5M)**	The datapath is implemented with all specifications provided in the problem statement but lacks innovative thinking **(4M)**	The datapath is moderately correct for the given specifications. **(3M)**	The datapath is implemented for only basic operations. **(2M)**	The datapath implementation is incomplete, without including any basic specification. **(1-0M)**	L3	CO 5	3.3.1

4.	Proficiency in Tool Usage **(5 M)**	Demonstrates expert-level proficiency, using the tool with complete confidence and precision.**(5 M)**	Demonstrates expert-level proficiency, using the tool with confidence.**(4 M)**	Demonstrates strong proficiency, with only minor issues or hesitations in tool usage **(3 M)**	Demonstrates proficiency, with only major issues in tool usage. **(2 M)**	Lacks the proficiency of tool usage. **(1 M)**	L3	CO 5	5.3.1
5.	Teamwork & collaboration. **(5 M)**	Excellent team collaboration and highly effective coordination. **(5M)**	Very good team collaboration and effective coordination. **(4M)**	Good team collaboration and adequate coordination. **(3M)**	Adequate team collaboration; generally, works well with some issues. **(2M)**	Poor team collaboration; lack of coordination and cohesion. **(1-0M)**	L3	CO 5	9.3.1

End Semester Assessment (ESA) Rubrics for Practical (P) component

ESA evaluation marks: 40

Sl. No.	Parameters	Excellent	Very Good	Good	Average	Scope for improvement	BL	CO	PI
1.	**Writeup-** Architecture & Design phase **(15M)**	Able to innovate in the design of the processor for the chosen instruction set **(15-13M)**	Able to design the processor for the chosen instruction set. **(12-10 M)**	Able to design the processor for few chosen instruction set. **(9-6M)**	Able to design the processor, but works for only few instructions with minor errors. **(5-3M)**	Unable to design. **(2-0M)**	L3	CO 3	3.3.1
2.	Simulation of Datapath **(15 M)**	Processor datapath is simulated successfully, and all outputs match expected results without any errors. **(15-13M)**	Processor datapath simulation produces mostly correct outputs with minor errors. **(12-10 M)**	Processor datapath simulation produces correct outputs for most test cases, but noticeable errors remain. **(9-6M)**	Processor datapath simulation shows errors in key outputs, though some parts function as expected. **(5-3M)**	Processor datapath simulation fails or produces incorrect results for most test cases. **(2-0M)**	L3	CO 5	5.3.1
3.	Performance analysis **(5 M)**	Able to compare the basic sequential processor design with provided advanced features in terms of several parameters.	Able to compare the basic sequential processor design with provided advanced	Lack of differences between the basic and advanced design. **(3M)**	Able to do only basic design. **(2M)**	Unable to identify the parameters to compare. **(1-0M)**	L3	CO 3	1.4.4

		(5M)	features in terms of few parameters. **(4M)**						
4.	Teamwork & collaboration. **(5 M)**	Excellent team collaboration and highly effective coordination. **(5M)**	Very good team collaboration and effective coordination. **(4M)**	Good team collaboration and adequate coordination. **(3M)**	Adequate team collaboration; generally, works well with some issues. **(2M)**	Poor team collaboration; lack of coordination and cohesion. **(1-0M)**	L3	CO 5	9.3.1

Course Unitization for ISA and ESA

		Theory			P Component		
Topics / Chapters	**Teaching Hours**	**No. of Questions in ISA-1**	**No. of Questions in ISA-2**	**No. of Questions in ESA**	**No. of Questions in ISA-1**	**No. of Questions in ISA-2**	**No. of Questions in ESA**
Unit I							
Chapter No. 1	04	1	-	1	-	-	-
Chapter No. 2	06	1	-	1	1	1	1
Chapter No. 3	06	1	-	1	1	1	1
Unit II							
Chapter No. 4	08	-	1.5	1.5	-	1	1
Chapter No. 5	08	-	1.5	1.5	-	1	1
Unit III							
Chapter No. 6	04	-	-	1	-	-	-
Chapter No. 7	04	-	-	1	-	-	-

Note

1. Each Question carries 20 marks and may also consists of sub-questions.
2. Mixing of sub-questions from different chapters within a unit (only for Unit I and Unit II) is allowed in ISA-I, II and ESA
3. Answer 5 full questions of 20 marks each (two full questions from Unit I, II and one full question from Unit III) out of 8 questions in ESA.

Date: **Head, SoCSE**

Course Assessment Plan

	Title: Computer Organization & Architecture				Course Code (21ECSC201)		
Course outcomes (COs)	**Weightage in assessment**					**Assessment Methods**	
	ISA					**ESA**	
	Weightage in assessment	**ISA-1**	**ISA-2**	**Practical (P) Component-1**	**Practical (P) Component-2**	**ESA**	**Practical (P) Component**
1. Acquire basic knowledge in computer fundamentals, performance analysis and interconnection principles in modern computing	13%	✓	--	✓	✓	✓	--
2. Build memory modules to perform I/O operations.	11%	✓	--	--	--	✓	--
3. Apply Instruction Set Architecture principles to effectively design and implement datapath in computer systems	38%	✓	✓	✓	✓	✓	✓
4. Explain the concept of parallelism with reference to computer performance.	26%	--	✓	--	--	✓	--

5. Integrate and simulate all modules to develop a complete processor, followed by a comprehensive performance analysis.	12%	--	--	✓	✓	--	✓
Weightage		16%	16%	8%	8%	42%	10%

Chapter-wise Plan

Course Code and Title: **Computer Organization and Architecture** (**21ECSC201**)	
Chapter Number and Title: 1. Computer Fundamentals	Planned Hours: 04

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain the general functions and structure of a digital computer.	CO 1	L2	1.4
2. Describe the key performance issues that relate to computer design	CO 1	L3	1.4
3. Interpret the basic elements of an instruction cycle & the role of interrupts.	CO 1	L3	1.4
4. Describe the concept of interconnection within a computer system.	CO 1	L2	1.4

Lesson Schedule Class No. - Portion covered per hour
1. Organization and Architecture, Structure and Function, History of Computers, The Evolution of the Intel x86 Architecture, Embedded Systems
2. Two Laws that Provide Insight: Amdahl's Law and Little's Law
3. Basic Measures of Computer Performance, Calculating the Mean, Benchmarks and Spec.
4. Computer Components, Computer Function, Interconnection Structures, Bus Interconnection, Point-to-Point Interconnect

Review Questions

Sl.No	Questions	TLOs	BL	PI Code
1.	What are the four main components of any general-purpose computer?	TLO 1	L2	1.4.4
2.	What, in general terms, is the distinction between i. computer organization and computer architecture ii. computer structure and computer function	TLO 1	L2	1.4.4
3.	A quad core processor could speed up a computer by a factor of 4 but this rarely happens. Use Amdahl's Law to compute the percentage of program execution that needs to be distributed across all 4 cores to achieve an overall speedup of i) 3 ii) 2 iii) 1.5 iv) 1.25.	TLO 2	L3	1.4.4

<table>
<tr><td>4.</td><td>Consider two different machines, with two different instruction sets, both of which have a clock rate of 200 MHz. The following measurements are recorded on the two machines running a given set of benchmark programs:
<table>
<tr><td>Instruction Type</td><td>Instruction Count (millions)</td><td>Cycles per Instruction</td></tr>
<tr><td>Machine A
Arithmetic and logic
Load and store
Branch
Others</td><td>8
4
2
4</td><td>1
3
4
3</td></tr>
<tr><td>Machine B
Arithmetic and logic
Load and store
Branch
Others</td><td>10
8
2
4</td><td>1
2
4
3</td></tr>
</table>
(i) Determine the effective CPI, MIPS rate and execution time for each machine.
(ii) Comment on the results.</td><td>TLO 2</td><td>L3</td><td>1.4.4</td></tr>
<tr><td>5.</td><td>Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields: the first byte contains the opcode and the remainder the immediate operand or an operand address.
a. What is the maximum directly addressable memory capacity (in bytes)?
b. How many bits are needed for the program counter and the instruction register?</td><td>TLO 3</td><td>L3</td><td>1.4.4</td></tr>
<tr><td>6.</td><td>What are interconnection structures? Explain different modules of interconnection structures with neat figures.</td><td>TLO 4</td><td>L3</td><td>1.4.4</td></tr>
</table>

Chapter-wise Plan

Course Code and Title: **Computer Organization and Architecture** (**21ECSC201**)	
Chapter Number and Title: 2. Computer System	Planned Hours: 06

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	**COs**	**BL**	**CA Code**
1. Describe the basic concepts, intent of cache memory and the key elements of cache design.	CO 2	L2	1.4
2. Distinguish among direct mapping, associative mapping, and set Associative mapping	CO 2	L3	1.4
3. Design larger memories using small memory modules.	CO 2	L3	1.4
4. Distinguish the types of semiconductor main memory.	CO 2	L3	1.4
5. Differentiate between programmed I/O, interrupt-driven I/O and direct memory access.	CO 2	L3	1.4

Lesson Schedule Class No. - Portion covered per hour
1. Computer Memory System Overview : Cache Memory Principles, Elements of Cache Design
2. Internal Memory: Semiconductor Main Memory,
3. Characteristics and Structure of DRAM
4. I/O Modules: External devices
5. Programmed I/O
6. Interrupt-Driven I/O& Direct Memory Access

Review Questions

Si.No.	Questions	TLOs	BL	PI Code
1.	How does the principle of locality relate to the use of multiple memory levels?	TLO 1	L3	1.4.4
2.	A set- associative cache consists of 64 lines, or slots, divided into four-line sets. Main memory contains 4K blocks of 128 words each. Show the format of main memory addresses.	TLO 2	L3	1.4.4
3.	For a direct- mapped cache, associative cache and set associative cache, main memory address is viewed as	TLO 2	L2	1.4.4

	consisting of three fields, two fields and three fields respectively. List and define the fields for each type of mapping.			
4.	Design and show the organization of memory chip of size 16M × 16 using 2048K × 8 memory chips. Find the total number of external connections.	TLO 3	L3	1.4.4
5.	What are the differences among sequential access, direct access, and random access?	TLO 4	L2	1.4.4
6.	How does SDRAM differ from ordinary DRAM	TLO 4	L2	1.4.4
7.	List and briefly define three techniques for performing I/O.	TLO 5	L2	1.4.4
8.	What is the difference between memory-mapped I/O and I/O-mapped I/O?	TLO 5	L2	1.4.4

Chapter-wise Plan

Course Code and Title: **Computer Organization and Architecture** (**21ECSC201**)	
Chapter Number and Title: 3. The Central Processing Unit	Planned Hours: **06**

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Present an overview of essential characteristics of machine instructions.	CO 3	L2	1.4
2. Describe the various types of addressing modes common in instruction sets.	CO 3	L3	1.4
3. Apply the knowledge of Instruction Set Architecture (ISA) to write and organize sequences of instructions.	CO 3	L3	2.1

Lesson Schedule Class No. - Portion covered per hour
1. Instruction Sets: Characteristics and Functions: Machine Instruction Characteristics
2. Types of Operands& Operations
3. Instruction Sets: Addressing Modes and Formats
4. Addressing Modes
5. Instruction Formats
6. Assembly Language

Review Questions

Sl.No.	Questions	TLOs	BL	PI Code
1.	What are the typical elements of a machine instruction?	TLO 1	L2	1.4.4
2.	List and briefly explain five important instruction set design issues.	TLO 1	L2	1.4.4
3.	What is the difference between an arithmetic shift and a logical shift?	TLO 2	L2	1.4.4
4.	Assume an instruction set that uses a fixed 16-bit instruction length. Operand specifiers are 6-bits in length. There are K two-operand instructions and L zero-operand instructions. What is the maximum number of one-operand instructions that can be supported?	TLO 2	L3	1.4.4

<table>
<tr><td>5.</td><td>Let the address stored in the program counter be designated by the symbol X1. Theinstruction stored in X1 has an address part (operand reference) X2. The operand needed to execute the instruction is stored in the memory word with address X3. An index register contains the value X4. What is the relationship between these various quantities if the addressing mode of the instruction is (a) direct; (b) indirect; (c) PC relative; (d) indexed?</td><td>TLO 2</td><td>L3</td><td>1.4.4</td></tr>
<tr><td>6.</td><td>In a computer system, a small part of memory is given in the following table.
<table>
<tr><th>Mem Address</th><th>Content</th></tr>
<tr><td>0x201</td><td>0X23AC</td></tr>
<tr><td>0x202</td><td>0X0205</td></tr>
<tr><td>0x203</td><td>0XAB44</td></tr>
<tr><td>0x204</td><td>0XC5A2</td></tr>
<tr><td>0x205</td><td>0X3012</td></tr>
<tr><td>0x206</td><td>0X27FF</td></tr>
</table>
What would be the contents of accumulator after running the following assembler code? (All values are in hexadecimal).
Assume initial content of R2 = 0x 2345
LOAD #0XA143
ROR
ADD [0X202]
AND #3002
SUB 0X0204
SHL
XOR R2
STORE 0x207</td><td>TLO 3</td><td>L3</td><td>2.1.2</td></tr>
</table>

Chapter-wise Plan

Course Code and Title: **Computer Organization and Architecture** (**21ECSC201**)	
Chapter Number and Title: 4. The Processor	Planned Hours:**08**

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Distinguish between user-visible and control/status registers, and discuss the purposes of registers in each category.	CO 3	L2	1.4
2. Describe the instruction cycle and instruction pipelining and associated pipeline hazards	CO 3	L3	1.4
3. Recognise the key characteristics of RISC machines and its implication for pipeline design and performance	CO 3	L3	2.1
4. Discuss the design issues involved in instruction level parallelism.	CO 4	L3	2.1

Lesson Schedule Class No. - Portion covered per hour
1. Processor Organization
2. Register Organization
3. Instruction Cycle
4. Instruction Cycle...cont
5. Instruction Pipelining
6. Instruction-Level Parallelism
7. Superscalar Processors: Overview
8. Design Issues

Review Questions

Sl.No.	Questions	TLOs	BL	PI Code
1.	What general roles are performed by processor registers?	TLO 1	L2	1.4.4
2.	If the last operation performed on a computer with an 8-bit word was an addition in which the two operands were 00000010 and 00000011, what would be the value of the following flags? a. Carry b. Zero c. Overflow d. Sign e. Even Parity f. Half-Carry	TLO 1	L3	1.4.4

3.	A microprocessor provides an instruction capable of moving a string of bytes from one area of memory to another. The fetching and initial decoding of the instruction takes 10 clock cycles. Thereafter, it takes 15 clock cycles to transfer each byte. The microprocessor is clocked at a rate of 10 GHz. a. Determine the length of the instruction cycle for the case of a string of 64 bytes. b. What is the worst-case delay for acknowledging an interrupt if the instruction is non-interruptible? c. Repeat part (b) assuming the instruction can be interrupted at the beginning of each byte transfer.	TLO 2	L3	1.4.4
4.	List and briefly explain various ways in which an instruction pipeline can deal with conditional branch instructions.	TLO 2	L2	1.4.4
5.	Identify some typical distinguishing characteristics of RISC organization?	TLO 3	L2	2.1.2
6.	Determine the distinction between instruction-level parallelism and machine parallelism?	TLO 4	L2	2.1.2
7.	Assume a superscalar pipeline is capable of fetching and decoding 2 instructions at a time - Instructions are fetched and decoded in pair. The next two instructions must wait to be decoded until the pair of decode pipeline stages has cleared. • having 3 separate ALUs (e.g., two for integer arithmetic and one for floating-point arithmetic) • 2 instances of the write-back pipeline stage • 6 instruction code fragment with the following constraints: - I1 requires two cycles to execute - I3 and I4 conflict for the same functional unit (e.g., both need floating-point arithmetic) - I5 depends on the value produced by I4 - I5 and I6 conflict for a functional unit • To fetch, decode, and write back an instruction, each stage need 1 clock cycle. • When there is a conflict for a functional unit, or when a functional unit requires more than one cycle to generate a result, instructions temporarily stall. Show the issue and completion policies for • In-order-issue and in-order completion • In-order-issue and out-of-order completion • Out-of-order -issue and out-of-order completion Comment on your results.	TLO 4	L3	2.1.2

Chapter-wise Plan

Course Code and Title: **Computer Organization and Architecture** (**21ECSC201**)	
Chapter Number and Title: 5. Control Unit Operation & Operating System support	Planned Hours: **08**

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	**COs**	**BL**	**CA Code**
1. Explain the concept of micro-operations and define the principal instruction cycle phases in terms of micro- operations	CO 3	L3	1.4
2. Discuss how micro- operations are organized to control a processor.	CO 3	L3	2.1
3. Summarize, at a top level, the key functions of an operating system.	CO 3	L2	1.4

Lesson Schedule Class No. - Portion covered per hour
1. Micro-Operations: Fetch cycle, Indirect cycle
2. Micro-Operations: Interrupt Cycle, Execute cycle, instruction cycle
3. Control of the Processor: Functional requirements
4. Control of the Processor: Control signals
5. Internal processor organization
6. Operating system overview
7. Scheduling
8. Memory management

Review Questions

Sl.No.	Questions	TLOs	BL	PI Code
1.	What are the basic functional elements of the processor? Explain their function.	TLO 1	L2	1.4.4
2.	What is the overall function of a processor’s control unit?	TLO 1	L2	1.4.4
3.	An ALU can add its two input registers, and it can logically complement the bits of either input register, but it cannot subtract. Numbers are to be stored in twos complement representation. List the micro- operations the control unit must perform to cause a subtraction.	TLO 2	L3	1.4.4
4.	List and briefly define the key services provided by an OS.	TLO 3	L2	1.4.4

5.	Briefly describe the four types of scheduling in OS	TLO 3	L2	1.4.4
6.	Identify the sequence of micro-operations and draw a neat diagram illustrating the fetch and indirect cycles for the given instructions. ADD R1, 0x30 STORE R1, (0x50)	TLO 2	L3	2.1.2

Chapter-wise Plan

Course Code and Title: **Computer Organization and Architecture** (**21ECSC201**)	
Chapter Number and Title: 6. Parallel Organization	Planned Hours: 04

Learning Outcomes: -

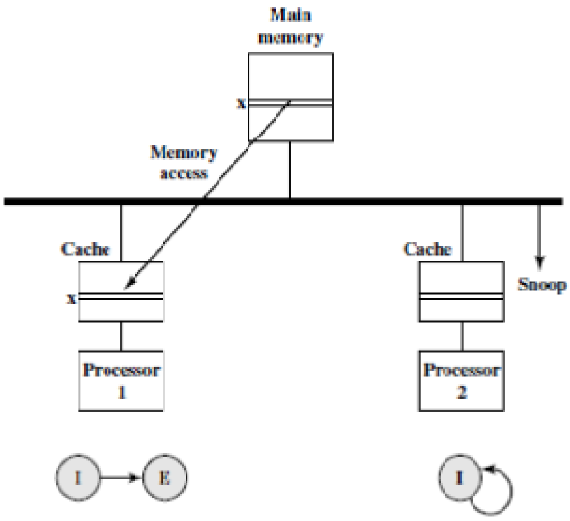
At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Summarize the types of parallel processor organizations and explain overview of design features of symmetric multiprocessors.	CO 4	L3	1.4
2. Explain the issue of cache coherence in a multiple processor system.	CO 4	L2	1.4
3. Identify methods for mitigating cache coherence.	CO 4	L3	2.1

Lesson Schedule Class No. - Portion covered per hour
1. Multiple Processor Organizations& Symmetric Multiprocessors
2. Cache Coherence , MESI Protocol
3. Multicore Computers: Hardware Performance Issues& Software Performance Issues
4. Multicore Organization& Heterogeneous Multicore Organization

Review Questions

Sl.No	Questions	TLOs	BL	PI Code
1.	What are some of the potential advantages of an SMP compared with a uniprocessor?	TLO 1	L2	1.4.4
2.	Let ‘a’ be the percentage of program code that can be executed simultaneously by n processors in a computer system. Assume that the remaining code must be executed sequentially by a single processor. Each processor has an execution rate of x MIPS. a. Derive an expression for the effective MIPS rate when using the system for exclusive execution of this program, in terms of n, a, and x. b. If n = 16 and x = 4 MIPS, determine the value of ‘a’, that will yield a system performance of 40 MIPS.	TLO 1	L3	1.4.4
3.	What are the beneficial features & the limitations of a bus organization?	TLO 2	L2	1.4.4
4.	Consider a situation in which two processors in an SMP configuration, over time, require access to the same line of data from main memory. Both processors have a cache and	TLO 3	L3	2.1.2

	use the MESI protocol. Initially, both caches have an invalid copy of the line. Figure depicts the consequence of a read of line x by Processor P1. If this is the start of a sequence of accesses, draw the subsequent figures for the following sequence: 1. P2 reads x. 2. P1 writes to x (for clarity, label the line in P1's cache x 3. P1 writes to x (label the line in P1's cache x 4. P2 reads x.  MESI Example: Processor 1 Reads Line x			

Chapter-wise Plan

Course Code and Title: **Computer Organization and Architecture** (**21ECSC201**)	
Chapter Number and Title: 7. General-Purpose Graphic Processing Units	Planned Hours: 04

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	**COs**	**BL**	**CA Code**
1. Present an overview of CUDA	CO 4	L2	1.4
2. Analyse the difference between a GPU and a CPU.	CO 4	L3	1.4
3. Describe the basic elements of a typical GPU architecture and analyse when to use a GPU as a coprocessor	CO 4	L3	1.4

Lesson Schedule Class No. - Portion covered per hour
1. Cuda Basics
2. GPU versus CPU
3. GPU Architecture Overview
4. GPU Architecture Overview...cont

Review Questions

Sl.No	Questions	TLOs	BL	PI Code
1.	List some examples of applications that benefit directly from the ability to scale throughput with the number of cores.	TLO 1	L3	1.4.4.
2.	Define CUDA.	TLO 2	L2	1.4.4.
3.	List the basic differences between CPU and GPU architectures	TLO 3	L3	1.4.4.
4.	What are the differences between kernel, thread, and block?	TLO 3	L2	1.4.4.

Model Question Paper for In Semester Assessment 1 (ISA-1)	
Course Code: 21ECSC201	**Course Title: Computer Organization and Architecture**
Duration: 75 minutes	**Max. Marks: 40**
Note: Answer two full questions Each full question carries equal marks.	

Q. No	Questions	Marks	CO	BL	PO	PI Code
1.a	Compare Programmed I/O and Interrupt driven I/O highlighting their advantages and limitations.	6	CO 2	L2	1.4	1.4.4
b	A quad core processor could speed up a computer by a factor of 4 but this rarely happens. Use Amdahl's Law to compute the percentage of program execution that needs to be distributed across all 4 cores to achieve an overall speedup of i) 3 ii) 2 iii) 1.5 iv) 1.25.	6	CO 1	L3	1.4	1.4.4
c	In a computer system, a small part of memory is given in the following table. <table><tr><th>Mem Address</th><th>Content</th></tr><tr><td>0x201</td><td>0X23AC</td></tr><tr><td>0x202</td><td>0X0205</td></tr><tr><td>0x203</td><td>0XAB44</td></tr><tr><td>0x204</td><td>0XC5A2</td></tr><tr><td>0x205</td><td>0X3012</td></tr><tr><td>0x206</td><td>0X27FF</td></tr></table>What would be the contents of accumulator after running the following assembler code? (All values are in hexadecimal). Assume initial content of R2 = 0x 2345 LOAD #0XA143 ROR ADD [0X202] AND #3002 SUB 0X0204 SHL XOR R2 STORE 0x207	8	CO 3	L3	2.1	2.1.2
2.a	Describe the parameters for consideration of design issues for the Instruction Set Architecture.	6	CO 3	L2	1.4	1.4.4
b	A main memory contains 32K blocks of 256 words each. A cache memory consists of 128 lines. Calculate the address field bits & draw the lines showing the mappings for each of the following Direct mapping 4-way set associative mapping	6	CO 2	L3	1.4	1.4.4
c	A bench mark program is run on a 40 MHz processor. The executed program consists of 100,000 instruction executions, with the following instruction mix and clock cycle count:	8	CO 1	L3	1.4	1.4.4

<table>
<tr>
<td></td>
<td>
<table>
<tr><th>Instruction type</th><th>Instruction count</th><th>Cycles per instruction</th></tr>
<tr><td>Integerarithmetic</td><td>45,000</td><td>1</td></tr>
<tr><td>Datatransfer</td><td>32,000</td><td>2</td></tr>
<tr><td>Floatingpoint</td><td>15,000</td><td>2</td></tr>
<tr><td>Controltransfer</td><td>8000</td><td>2</td></tr>
</table>
Determine the effective CPI, MIPS rate and execution time for this program.
</td>
<td></td><td></td><td></td><td></td><td></td>
</tr>
<tr>
<td>3.a</td>
<td>Explain instruction cycle state diagram with interrupts.</td>
<td>6</td><td>CO 1</td><td>L2</td><td>1.4</td><td>1.4.4</td>
</tr>
<tr>
<td>b</td>
<td>Using i) 2 address format
ii) 3 address format
Write assembly code for the instruction: $D = \frac{(A-3B)(2C)}{B+C}$ where A,B,C and D are addresses of locations in memory.</td>
<td>6</td><td>CO 3</td><td>L3</td><td>2.1</td><td>2.1.2</td>
</tr>
<tr>
<td>c</td>
<td>Design a 16M x 32 size memory chip using memory chips of
1024K x 16
2048K x 8
Draw the diagrams showing proper connections for each of the above cases and list out the required external connections.</td>
<td>8</td><td>CO 2</td><td>L3</td><td>1.4</td><td>1.4.4</td>
</tr>
</table>

Model Question Paper for In Semester Assessment 2 (ISA-2)	
Course Code: 21ECSC201	**Course Title: Computer Organization and Architecture**
Duration: 75 minutes	**Max. Marks: 40**
Note: Answer two full questions Each full question carries equal marks.	

Q.No	Questions	Marks	CO	BL	PO	PI Code
1.a (ch-4)	Explain various branch prediction techniques and with flow chart describe dynamic branch prediction.	10	CO 4	L2	1.4	1.4.4
B (Ch-5)	Write a set of microinstructions for the instructions given below ADD R1, 0x30 STORE R1, (0x50) With a neat diagram of fetch and indirect cycle explain the data flow for the above instructions.	10	CO 3	L3	1.4	1.4.4
2.a (Ch-4)	Assume a superscalar pipeline with operand forwarding capable of fetching and decoding two instructions at a time, having three separate execution units and having two instances of the write-back units. Consider the following instruction fragment with 8 instructions I1…I8: • I1, I2 requires two cycles to execute. • I3 depends on I2 • I5 requires two cycles to execute • I5 and I6 compete for the same execution unit • I7 depends on I6 Show the issue and completion policies for i. In-order issue and out-of-order completion ii. Out-of-order issue and out-of-order completion.	10	CO 4	L3	2.1	2.1.2
b. (Ch-5)	i. What are the rules for micrograping? Write a set of microinstructions for the instruction AC ← (AC) + (Memory) applying above rules on an internal CPU bus. ii. Consider the instruction cycle code for phases Fetch, Indirect, Execute and interrupt as **Fetch Phase:00, Indirect:01, Execute: 10 and Interrupt: 11** Show the complete execution phase for the instruction Mul R1, (0x40) using the flowchart for the instruction cycle code.	10	CO 3	L3	2.1	2.1.2
3.b (Ch-5)	i. What are micro-operations? Explain the rules for micro-grouping. ii. Write micro-instruction for fetch-indirect addressing cycle.	10	CO 3	L2	1.4	1.4.4

Model Question Paper for End Semester Assessment (ESA)	
Course Code: 21ECSC201	**Course Title: Computer Organization and Architecture**
Duration: 03 hrs	**Max. Marks: 100 marks**

Note: Answer any five questions selecting two questions from UNIT-1,UNIT-2 and one Question from UNIT-3. Each full question carries equal marks.

UNIT-1

Q.No	Questions	Marks	CO	BL	PO	PI Code
1.a	Compare Programmed I/O and Interrupt driven I/O highlighting their advantages and limitations.	6	CO 2	L2	1.4	1.4.4
b	A quad core processor could speed up a computer by a factor of 4 but this rarely happens. Use Amdahl's Law to compute the percentage of program execution that needs to be distributed across all 4 cores to achieve an overall speedup of i) 3 ii) 2 iii) 1.5 iv) 1.25.	6	CO 1	L3	1.4	1.4.4
c	In a computer system, a small part of memory is given in the following table. <table><tr><th>Mem Address</th><th>Content</th></tr><tr><td>0x201</td><td>0X23AC</td></tr><tr><td>0x202</td><td>0X0205</td></tr><tr><td>0x203</td><td>0XAB44</td></tr><tr><td>0x204</td><td>0XC5A2</td></tr><tr><td>0x205</td><td>0X3012</td></tr><tr><td>0x206</td><td>0X27FF</td></tr></table>What would be the contents of accumulator after running the following assembler code? (All values are in hexadecimal). Assume initial content of R2 = 0x 2345 LOAD #0XA143 ROR ADD [0X202] AND #3002 SUB 0X0204 SHL XOR R2 STORE 0x207	8	CO 3	L3	2.1	2.1.2
2.a	Describe the parameters for consideration of design issues for the Instruction Set Architecture.	6	CO 3	L2	1.4	1.4.4
b	A main memory contains 32K blocks of 256 words each. A cache memory consists of 128 lines. Calculate the address field bits & draw the lines showing the mappings for each of the following Direct mapping 4-way set associative mapping	6	CO 2	L3	1.4	1.4.4
c	A bench mark program is run on a 40 MHz processor. The	8	CO 1	L3	1.4	1.4.4

	executed program consists of 100,000 instruction executions, with the following instruction mix and clock cycle count: **Instruction type** / **Instruction count** / **Cycles per instruction** Integerarithmetic / 45,000 / 1 Datatransfer / 32,000 / 2 Floatingpoint / 15,000 / 2 Controltransfer / 8000 / 2 Determine the effective CPI, MIPS rate and execution time for this program.					
3.a	Explain instruction cycle state diagram with interrupts.	6	CO 1	L2	1.4	1.4.4
b	Using i) 2 address format ii) 3 address format Write assembly code for the instruction: $D = \frac{(A-3B)(2C)}{B+C}$ where A,B,C and D are addresses of locations in memory.	6	CO 3	L3	2.1	2.1.2
c	Design a 16M x 32 size memory chip using memory chips of 1024K x 16 2048K x 8 Draw the diagrams showing proper connections for each of the above cases and list out the required external connections.	8	CO 2	L3	1.4	1.4.4
	UNIT-2					
4.a	Explain various branch prediction techniques and with flow chart describe dynamic branch prediction.	10	CO 4	L2	1.4	1.4.4
b	Write a set of microinstructions for the instructions given below ADD R1, 0x30 STORE R1, (0x50) With a neat diagram of fetch and indirect cycle explain the data flow for the above instructions.	10	CO 3	L3	1.4	1.4.4
5.a	Assume a superscalar pipeline with operand forwarding capable of fetching and decoding two instructions at a time, having three separate execution units and having two instances of the write-back units. Consider the following instruction fragment with 8 instructions I1…I8: • I1, I2 requires two cycles to execute. • I3 depends on I2 • I5 requires two cycles to execute • I5 and I6 compete for the same execution unit • I7 depends on I6 Show the issue and completion policies for iii. In-order issue and out-of-order completion iv. Out-of-order issue and out-of-order completion.	10	CO 4	L3	2.1	2.1.2
b	iii. What are the rules for microgrouping? Write a set of	10	CO 3	L3	2.1	2.1.2

The nested instruction mix table:

Instruction type	Instruction count	Cycles per instruction
Integerarithmetic	45,000	1
Datatransfer	32,000	2
Floatingpoint	15,000	2
Controltransfer	8000	2

	microinstructions for the instruction AC ← (AC) + (Memory) applying above rules on an internal CPU bus. iv. Consider the instruction cycle code for phases Fetch, Indirect, Execute and interrupt as **Fetch Phase:00, Indirect:01, Execute: 10 and Interrupt: 11** Show the complete execution phase for the instruction Mul R1, (0x40) using the flowchart for the instruction cycle code.					
6.a	iii. What are micro-operations? Explain the rules for micro-grouping. iv. Write micro-instruction for fetch-indirect addressing cycle.	10	CO 3	L2	1.4	1.4.4
b	Explain various branch prediction techniques and with flow chart describe dynamic branch prediction.	10	CO 4	L2	1.4	1.4.4
UNIT-3						
7.a.	Explain symmetric multiprocessor organization with a neat block diagram. State the differences between multiprogramming and multiprocessing.	10	CO 4	L2	1.4	1.4.4
b.	Compare & contrast write invalidate and write update w.r.to snoopy protocol. Consider a situation in which there are 5 SMPs P1, P2, P3 P4 and P5. All of them have a cache and use the MESI protocol. Initially, P1 is in the exclusive state w.r.t line x. Following are the sequence of events i) P3 intends to modify x. ii) P5 wants to read x iii) P4 intends to modify x With a neat state diagram explain transitions that take place for this situation.	10	CO 4	L3	1.4	1.4.4
8.a.	Describe the three general sections of a CUDA program. What are the basic differences between CPU and GPU architectures?	10	CO 4	L2	1.4	1.4.4
b.	Describe the salient features of the attributes of GPU's memory hierarchy.	10	CO 4	L3	1.4	1.4.4